

Training Leaves Traces: White-Box Model Lineage Verification using Diagonal-Dominant Fingerprints

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Abstract

Open-weight checkpoints are routinely fine-tuned, quantized, pruned, merged, and redistributed without reliable lineage metadata. Existing provenance tools are brittle: hashes fail after any weight change, metadata requires trust, and watermarks must be inserted before release. **We study a post-hoc white-box weight-lineage verification: given two checkpoints with compatible residual architecture, do they share weight-level lineage after common checkpoint transformations?** This is distinct from behavioral similarity: a distilled student imitates its teacher’s outputs, but its weights are trained from scratch and share no lineage.

We show that residual-network training induces an intrinsic fingerprint: correctly paired residual-branch products develop a diagonal-dominant fingerprint, measured by a trace/Frobenius score, that is absent at initialization and in non-residual controls. We use this signal to align blocks between checkpoints and compute a model-level lineage score from centered branch signatures. Across 12 architectures spanning language, vision, and speech (GPT-2 to LLaMA-2, ViT, Whisper, ResNets), the fingerprint recovers correct projection pairings with 80–100% accuracy, and the lineage score achieves complete separation: fine-tuned, pruned, quantized, and LoRA-merged checkpoints score high, while independently trained models and distilled students remain low. Unlike activation-based CKA and IPGuard, which track functional similarity, this provides a passive, retroactive provenance signal for white-box residual checkpoints.

1 Introduction

The open-weight ecosystem has transformed how neural networks are distributed and reused. When Meta released LLaMA (Touvron et al. 2023), the community produced instruction-tuned variants within weeks. Platforms like Hugging Face now host hundreds of thousands of model checkpoints (Hugging Face 2024), many of which are fine-tuned (Devlin et al. 2019; Ouyang et al. 2022), quantized (Han, Mao, and Dally 2016; Dettmers et al. 2022), pruned (Han et al. 2015), or merged (Wortsman et al. 2022; Ilharco et al. 2023). This creates a *model supply chain* where downstream checkpoints inherit weights from upstream models, but the lineage is often undocumented or intentionally obscured. Provenance of AI-generated content is now treated as a first-class problem: Google’s SynthID watermarks text and images at generation time (Dathathri et al.

2024), yet no analogous mechanism exists for the models themselves. Model reuse is easy; reliable model provenance is not.

Existing provenance mechanisms fall short (Table 1). Cryptographic hashes break after any weight modification. Metadata and model cards (Mitchell et al. 2019) require honest distributors and preserved records; they can be falsified or lost. Watermarks (Uchida et al. 2017; Adi et al. 2018) must be inserted before release; they cannot be applied retroactively to already-deployed checkpoints, and no surveyed scheme proved robust against fine-tuning or pruning (Lukas et al. 2022). Behavioral tests that compare model outputs cannot distinguish weight inheritance from imitation: two models can behave similarly without sharing weights, especially after distillation (Hinton, Vinyals, and Dean 2015). Activation-based similarity measures like CKA (Kornblith et al. 2019) and decision-boundary fingerprints like IPGuard (Cao, Jia, and Gong 2021) inherit this limitation. We need a passive, post-hoc signal that reads provenance from the weights themselves.

We ask: given a reference checkpoint and a suspect checkpoint with the same residual architecture (e.g., both LLaMA-2-7B derivatives), can we determine whether the suspect inherited weights from the reference, even after common transformations like fine-tuning, quantization, or pruning? This is distinct from duplicate detection (which hashes solve until the first modification), behavioral similarity (which conflates imitation with inheritance), and watermark verification (which requires forethought). The question is whether shared weight ancestry leaves detectable structure after modification.

The answer is yes. We find that training itself leaves a characteristic fingerprint in the weights of residual networks. In these architectures (He et al. 2016), each block adds a learned transformation to its input via a skip connection (Figure 1a). During training, the input and output projections of each block align so that their product $W_{\text{out}}W_{\text{in}}$ develops diagonal-dominant structure that is absent at random initialization and does not appear in networks without skip connections (Figure 1b). This structure is a product of gradient coupling: because W_{in} and W_{out} sit in the same residual branch, every backward pass sends correlated updates through both, and the correlation accumulates into diagonal structure in their product. The fingerprint is absent at initial-

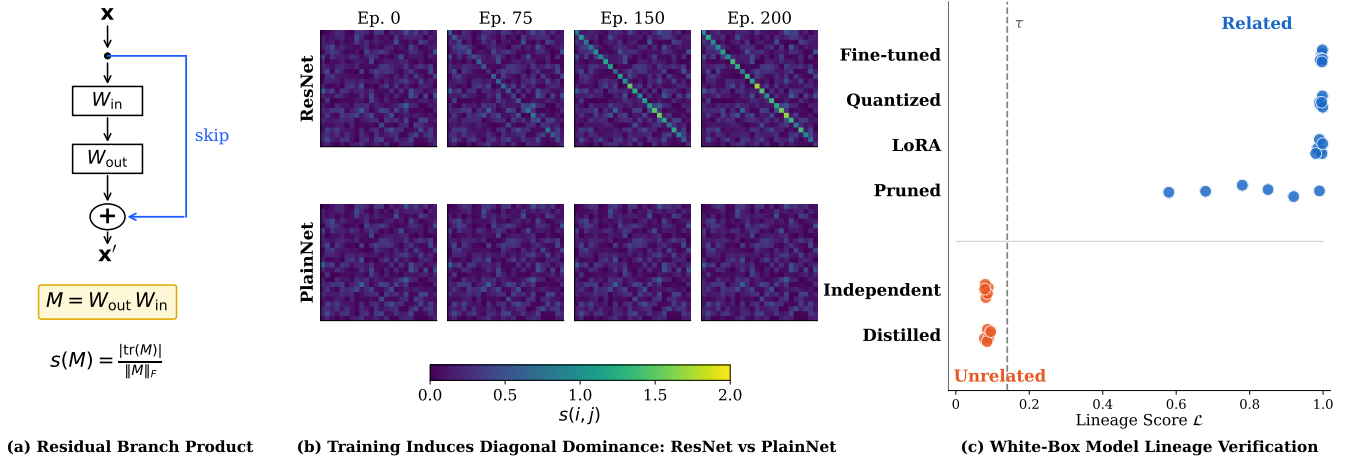


Figure 1: Training leaves traces in residual networks. (a) The residual branch product $M = W_{\text{out}}W_{\text{in}}$ for a block with skip connection; the diagonal-dominance score $s(M)$ measures trace concentration. (b) Training dynamics: score matrices $s(i, j)$ from epoch 0 to 200. ResNet (top row) develops strong diagonal structure (100% block-pairing accuracy); PlainNet without skip connections (bottom row) remains random (3% accuracy). The fingerprint is learned, not an initialization artifact. (c) Lineage verification: the lineage score $\mathcal{L}$ (Eq. 7) aggregates per-block signature similarity. Related checkpoints (fine-tuned, quantized, pruned, LoRA-merged) score high; unrelated checkpoints (independent, distilled) score low, enabling clean separation.

ization, never appears without the skip connection, and can be destroyed or created by manipulating the coupling alone (Section 5.1).

We exploit this fingerprint for provenance verification. Each residual block yields a signature: its branch product scored for diagonal concentration via a trace/Frobenius ratio (Equation 3). Correctly paired blocks from the same trained model score high, while mismatched or unrelated blocks do not. The extraction is architecture-aware, respecting the specific structure of Transformer MLPs, attention paths, and ResNet bottleneck blocks (Table 2). To compare two checkpoints, we align their blocks using these signatures, aggregate the pairwise similarities into a model-level lineage score, and threshold it against a null distribution built from unrelated models (Equation 7).

The fingerprint is not specific to one model family. We observe it in public checkpoints across language, vision, and speech (GPT-2, BERT, LLaMA-2, Mistral, Qwen2.5, DeepSeek-R1, ViT, Whisper, and ResNets) where it recovers block correspondences with over 91% accuracy (Table 3). Turning to the verification task itself, we construct controlled lineage benchmarks on MLPs and ResNet-18: models with genuine weight ancestry (produced by fine-tuning, weight noise, pruning, quantization, or LoRA merging) are cleanly separated from those without (independent training runs and distilled students), with AUROC = 1.000. The distillation case is instructive: a student can closely replicate its teacher’s outputs, yet our score correctly labels it unrelated, because its weights carry no trace of the teacher’s training (Figure 1c). Finally, on the public LLaMA-2 family, the method flags three real fine-tuned descendants (chat, Vicuna, CodeLlama) as related while rejecting four independently trained checkpoints, among them OpenLLaMA, which shares the architecture but inherits no weights.

Our contributions:

- **Diagonal-dominant fingerprint.** We establish that gradient-based training drives the input and output projections of residual blocks toward mutual alignment, concentrating their product along the diagonal, a property absent at random initialization and never acquired by skip-free architectures.
- **White-box verification method.** We convert this signature into a practical test: per-block scores are matched across two checkpoints, aggregated into a single lineage statistic, and judged against a null distribution, requiring no watermark, no metadata, and no cooperation from the model’s distributor.
- **Broad empirical validation.** We evaluate on 246 checkpoint pairs covering 12 architectures and 5 model families, from synthetic benchmarks with known ground truth to real-world derivatives of LLaMA-2, obtaining 80–100% block correspondence and AUROC=1.000.
- **Robustness and specificity.** The fingerprint persists through fine-tuning, quantization, pruning, and LoRA merging, yet vanishes for distilled students and independently trained models of identical architecture, a distinction beyond the reach of behavioral measures such as CKA and IPGuard.

2 Problem Setting and Related Work

2.1 White-Box Model Lineage Verification

A verifier receives two checkpoints A and B and determines whether they share weight-level lineage. We assume white-box access and compatible residual architectures (same L and d); cross-architecture comparisons are out of scope. The verifier operates on weights alone: no training data, activations, or forward passes.

Method	Retro.	FT	Quant	Distill
Crypto Hash	✓	✗	✗	✓
Metadata/Logs	✓	✓	✓	✓
Watermarking	✗	~	~	✗
Behavioral Fingerprints	✓	✓	✓	✗
Ours	✓	✓	✓	✓

Table 1: Comparison with prior provenance methods. Retro.: applicable retroactively. FT: survives fine-tuning. Quant: survives quantization. Distill: correctly classifies distilled copies as unrelated. ~: partial or unreliable.

Checkpoints are *related* if they share a common weight ancestor through transformations $\theta' = T(\theta, D)$: fine-tuning, RLHF, pruning, quantization, or LoRA merging. They are *unrelated* if initialized independently, even when architecture, data, task, or outputs match. Linear merges induce partial lineage (Appendix C.6). The verifier returns RELATED, UNRELATED, or INCONCLUSIVE. The score is symmetric, $\mathcal{L}(A, B) = \mathcal{L}(B, A)$; directional attribution requires metadata (Appendix B). Block-level correspondences are recovered as a diagnostic (Section 4.1).

Lineage follows initialization, not behavior. A distilled student trained from fresh weights is unrelated to its teacher, regardless of output fidelity; model-extraction copies via query access (Tramèr et al. 2016) are likewise unrelated. Conversely, a student initialized from teacher weights is related regardless of the training objective. The method is a forensic primitive for weight-level provenance, complementing cryptographic signing and watermarking where those were never applied.

2.2 Related Work

Table 1 summarizes the comparison.

Cryptographic hashes break after any weight modification. Model cards (Mitchell et al. 2019) require honest distributors and can be falsified when checkpoints are redistributed. Proof-of-learning (Jia et al. 2021) requires retained training transcripts that already-released checkpoints lack.

Parameter watermarks (Uchida et al. 2017) and backdoor watermarks (Adi et al. 2018) embed deliberate signals during training. Lukas et al. (2022) found no surveyed scheme that reliably survived fine-tuning and pruning, and all require proactive insertion: unwatermarked legacy checkpoints cannot be verified.

IPGuard (Cao, Jia, and Gong 2021) extracts inputs near classification boundaries; adversarial frontier stitching (Le Merrer, Perez, and Trédan 2020) marks decision surfaces with adversarial examples. CKA (Kornblith et al. 2019) and SVCCA (Raghu et al. 2017) compare representations on a probe dataset. All four measure functional similarity and cannot distinguish inheritance from imitation: a distilled student may share decision surfaces while having independent weights. They also require data and forward passes.

Architecture	Residual branch $F(x)$	Product M
Transformer MLP / BasicBlock	$W_2\sigma(W_1x)$	W_2W_1
Attention V/O path	$W_O\text{Attn}(x)W_Vx$	W_OW_V
Attention Q/K path	bilinear $x^\top W_QW_K^\top x$	$W_QW_K^\top$
Bottleneck ResNet	$W_3\sigma(W_2\sigma(W_1x))$	$W_3W_2W_1$
SwiGLU MLP	$W_{\text{down}}[\sigma(W_{\text{gate}}x) \odot W_{\text{up}}x]$	$W_{\text{down}}W_{\text{up}}$

Table 2: Architecture-aware residual branch products. Each M composes the linear maps of one branch (dropping nonlinearities) so it maps the residual stream to itself. Factorization must match the architecture; incomplete products destroy the signal (Section 5).

Weight cosine, Frobenius distance, and permutation-aligned matching (Ainsworth, Hayase, and Srinivasa 2023) detect derived checkpoints when the suspect remains close in parameter space, but these are uncalibrated global scalars with no structural account of lineage. HuRef (Zeng et al. 2024) relies on LLM parameter convergence after pretraining. REEF (Zhang et al. 2024) and neural lineage detection (Yu et al. 2024) need probe data or activations. MoTher (Horwitz and Hoshen 2025) recovers model trees but requires a checkpoint collection rather than a single pair. Our fingerprint is data-free, operates on a single pair against a calibrated null, and localizes inheritance to specific blocks.

3 Training-Induced Diagonal Dominance in Residual Branch Products

A residual block computes $x_{\ell+1} = x_\ell + F_\ell(x_\ell)$, where F_ℓ is the residual branch. For a branch with K linear layers $W_1, \dots, W_K$ interleaved with nonlinearities, the *residual branch product* composes the linear factors, dropping nonlinearities:

$$M_\ell = W_K W_{K-1} \cdots W_1 \in \mathbb{R}^{d \times d} \quad (1)$$

Individual factors W_k are rectangular and do not share an input–output basis; their composed product maps the residual stream back into its own coordinate space, so its diagonal entries compare like with like. The exact factorization is architecture-dependent (Table 2).

3.1 Why Residual Training Induces Diagonal Dominance

Under the Frobenius inner product $\langle A, B \rangle = \text{tr}(A^\top B)$, the identity matrix and the traceless matrices span orthogonal subspaces of $\mathbb{R}^{d \times d}$. Any branch product therefore splits into a component along the identity, with coefficient $\langle M_\ell, I \rangle / \langle I, I \rangle = \text{tr}(M_\ell)/d$, and an orthogonal traceless remainder:

$$M_\ell = -\varepsilon_\ell I + E_\ell \quad (2)$$

where $\varepsilon_\ell = -\text{tr}(M_\ell)/d$ and $\text{tr}(E_\ell) = 0$. Traces are predominantly negative (Section 5), so ε_ℓ is typically positive; the score uses magnitude.

The *diagonal-dominance score* measures this concentration:

$$s(M) = \frac{|\text{tr}(M)|}{\|M\|_F}, \quad (3)$$

the fraction of the matrix’s energy along the identity direction, up to $\sqrt{d}$. Training moves energy into the identity component: correctly paired branch products score far above unpaired or untrained ones (Figure 1b).

The skip connection changes what the branch must learn. A plain layer must transmit the entire representation; a residual branch learns only a correction around an identity path that already transmits everything. This has a gradient consequence: W_{in} and W_{out} are the two ends of one correction, and the loss reaches both through the same branch. ∇W_{out} depends on the branch input, a function of W_{in} ; ∇W_{in} depends on the upstream gradient, which flows through W_{out} . Their updates are therefore correlated at every step by construction. In a plain network no such pairing is privileged—every adjacent pair of layers is coupled equally—so no per-block signature can form.

Why does the correlation accumulate along the identity direction specifically? The skip places the branch in an identity-centered parameterization: training optimizes a small correction to a near-identity map, and the linear component of coordinated small corrections lies along coordinate-preserving directions. The decomposition of Equation (2) isolates exactly this component; Section 5.1 shows that supplying it directly, with no backpropagation, suffices to build the fingerprint.

A natural alternative account is dynamical isometry (Pennington, Schoenholz, and Ganguli 2017): if training enforced near-orthogonal block Jacobians $J = I + J_F$, diagonal dominance would follow as a byproduct of pressure toward $J \approx I$. This account makes a testable prediction—trained Jacobians should be more orthogonal than at initialization—which Section 5.1 falsifies directly. We therefore present gradient coupling, not isometry, as the mechanism.

3.2 Margin Analysis

By orthogonality of decomposition (2), $\|M_\ell\|_F^2 = \varepsilon_\ell^2 d + \|E_\ell\|_F^2$: the score depends on M only through the ratio of identity to traceless components:

Proposition 1 (Correct-pair score). *Let $M = -\varepsilon I + E$ with $\text{tr}(E) = 0$ and $\varepsilon \neq 0$. Then:*

$$s(M) = \frac{\sqrt{d}}{\sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}} \leq \sqrt{d} \quad (4)$$

with equality if and only if $E = 0$.

Proof. $|\text{tr}(M)| = |\varepsilon|d$ and $\|M\|_F^2 = \varepsilon^2 d + \|E\|_F^2$ by orthogonality. Substitute into (3).

Geometrically, $s(M)/\sqrt{d}$ is the cosine of the angle between M and the identity. Mismatched products have no such concentration: W_{out} and W_{in} from different blocks have no preferred alignment, giving

Corollary 1 (Mismatched baseline). *For products of projections from independent blocks, $\mathbb{E}[s] \approx 1/\sqrt{d}$.*

The margin scales as d : wider networks separate correct from incorrect pairings more cleanly. Empirically the

correct-pair score grows as $d^{0.87}$ (Section 5), consistent with $\|E\|_F/(\varepsilon\sqrt{d})$ shrinking as d grows while remaining below the $\sqrt{d}$ bound.

4 From Diagonal-Dominance Fingerprints to Lineage Verification

Diagonal dominance certifies trained residual structure, not ancestry: unrelated trained models both exhibit it. The decomposition (2) separates the two signals. The identity component is generic across trained models. The traceless component E_ℓ is checkpoint-specific: it survives weight-preserving transformations but not reinitialization. Verification runs on E_ℓ , with diagonal dominance gating which branches are trustworthy.

4.1 Centered Residual Signatures and Block Alignment

For each branch product M_ℓ we retain only the traceless remainder of decomposition (2), normalized to a unit vector:

$$R_\ell = M_\ell - \frac{\text{tr}(M_\ell)}{d} \cdot I, \quad \phi_\ell = \frac{\text{vec}(R_\ell)}{\|\text{vec}(R_\ell)\|_2} \quad (5)$$

$R_\ell = E_\ell$ from (2). The signature ϕ_ℓ carries checkpoint-specific geometry that weight-preserving transformations inherit and independent training cannot reproduce. Because M_ℓ is invariant under hidden permutation and reciprocal rescaling, so is every signature built on it (Section 5.6).

Given reference A and suspect B with L blocks each, we form the similarity matrix $G_{ij} = \langle \phi_i^A, \phi_j^B \rangle$, gated by diagonal dominance (Appendix B), and recover the block correspondence via the Hungarian algorithm (Kuhn 1955):

$$\pi^* = \arg \max_{\pi \in \mathcal{P}_L} \sum_i G_{i, \pi(i)} \quad (6)$$

For derived checkpoints preserving block order, π^* should be the identity; recovering it without metadata confirms that signatures track block identity. The correspondence matrix also shows which blocks were inherited (Appendix C). Complexity: branch products $O(Ld^2h)$, similarity matrix $O(L^2d^2)$, assignment $O(L^3)$.

4.2 Lineage Score and Verification Test

The lineage score averages signature similarity over aligned branches:

$$\mathcal{L}(A, B) = \frac{1}{L} \sum_{\ell=1}^L \langle \phi_\ell^A, \phi_{\pi^*(\ell)}^B \rangle \quad (7)$$

Each term is a cosine, so $\mathcal{L} \in [-1, 1]$ with $\mathcal{L}(A, A) = 1$. The score is symmetric (Appendix B). Related checkpoints score near 1; unrelated ones near 0, since independently initialized weights produce uncorrelated E_ℓ .

We calibrate against a null distribution $\mathcal{N}_A = \{\mathcal{L}(A, U_j)\}$ from independently trained models U_j , with mean μ_0 and standard deviation σ_0 :

$$Z(A, B) = \frac{\mathcal{L}(A, B) - \mu_0}{\sigma_0} \quad (8)$$

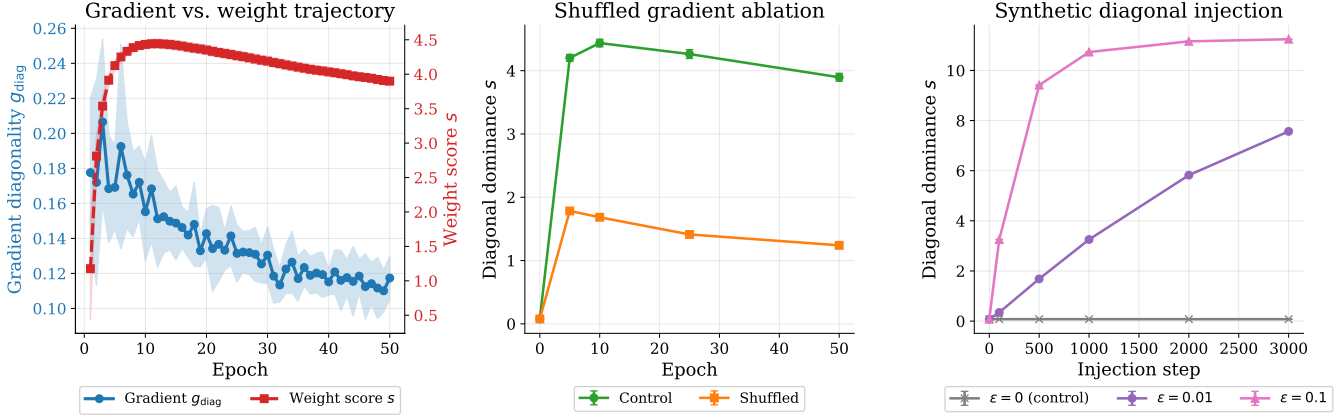


Figure 2: Gradient coupling builds the diagonal fingerprint. (a) The gradient product itself does not become diagonal, so the mechanism couples update directions, not diagonal shape. (b) Necessity: shuffling ∇W_{out} across blocks drops the score by 68% while training proceeds. (c) Sufficiency: direct diagonal injection builds the score with no backpropagation.

The protocol returns

$$V(A, B) = \begin{cases} \text{RELATED} & \text{if } Z > \tau_{\text{upper}} \\ \text{UNRELATED} & \text{if } Z < \tau_{\text{lower}} \\ \text{INCONCLUSIVE} & \text{otherwise} \end{cases} \quad (9)$$

Thresholds $\tau_{\text{upper}} = 3.0$, $\tau_{\text{lower}} = -3.0$ give false-positive rate below 0.13% under a Gaussian null. Compatibility (same L and d) is checked before scoring; the protocol returns INCOMPATIBLE otherwise. Memory: $O(L^2 + d^2)$ beyond the checkpoints.

5 Experiments

We validate the claims of Sections 3–4: fingerprint existence (Section 5.1), cross-architecture generalization (Section 5.2), lineage verification (Section 5.3), and baseline comparisons (Section 5.4).

5.1 The Fingerprint Exists and Is Learned

Section 3.1 predicts that the fingerprint is learned, requires the skip connection, and arises from gradient coupling rather than isometry. We test all three.

We train depth-24 residual MLPs (in_dim=24, hidden=64) on CIFAR-10 for 200 epochs (3 seeds) under seven initialization schemes: orthogonal, Kaiming-normal/uniform, Xavier-normal/uniform, uniform, and Gaussian $\sigma=0.02$. Pair accuracy starts at chance ($\leq 2.1\%$) and reaches 93–100% after training for every scheme that converges (Appendix C; Gaussian $\sigma=0.02$ collapses blocks). Orthogonal initialization satisfies dynamical isometry exactly at initialization yet yields zero correct pairs. Isometry without training produces nothing.

Using the same architecture, we compare ResNet-24 against PlainNet-24, which removes only the skip connection. ResNet-24 reaches 100% pair accuracy (AUROC 0.98); PlainNet-24, trained to similar loss (1.56 vs. 1.35), stays at chance (3%, AUROC 0.51). Training alone does not produce the fingerprint; the skip connection is required.

Then, we measure the normalized Jacobian deviation $\delta_J^{\text{norm}} = \|J^\top J / \|J\|_F^2 - I/d\|_F$ across the GPT-2 family (small/medium/large/XL), comparing pretrained weights against random initialization. Trained GPT-2 Jacobians are 5–12 $\times$ less orthogonal than at initialization (GPT-2-small: 0.297 vs 0.025). The fingerprint grows while orthogonality decays, ruling out isometry (Appendix C.2).

During training of 48-block residual MLPs (in_dim=128, hidden=256, CIFAR-10, 50 epochs, 3 seeds), we track whether the gradients themselves are diagonal ($g_{\text{diag}} = |\text{tr}(\nabla W_{\text{out}} \nabla W_{\text{in}})| / \|\cdot\|_F$) and whether the weights become diagonal ($s(M)$). The gradient product stays flat (~ 0.15), meaning individual updates are not diagonal. Yet the weight score climbs from 0.08 to ~ 4.0 (Figure 2a). The diagonal structure emerges from many small correlated updates accumulating over training, not from any single diagonal gradient step.

Two interventions isolate gradient coupling as the cause. *Necessity*: shuffling ∇W_{out} across blocks before each optimizer step (block i 's W_{out} receives gradients from a random block j) breaks only the within-block pairing while preserving valid gradients. This cuts the fingerprint by 68% ($s: 3.90 \rightarrow 1.24$) at comparable accuracy (47% vs 54%) (Figure 2b). *Sufficiency*: injecting synthetic updates $\Delta W_{\text{in}} = -\varepsilon \cdot e_i$, $\Delta W_{\text{out}} = -\varepsilon \cdot e_i^\top$ ($\varepsilon=0.01$) directly into weights for 3000 steps, with no loss function or backpropagation, builds the fingerprint from scratch ($s: 0.08 \rightarrow 7.57$) (Figure 2c). Removing the coupling removes the fingerprint; supplying the coupling alone creates it.

Training on CIFAR-10 with randomly shuffled labels (100 epochs, 3 seeds) produces a stronger fingerprint than normal training ($s=5.88$ vs. 3.42), even though networks can only memorize ($\sim 10\%$ test accuracy). The fingerprint tracks optimization dynamics, not task learning.

5.2 The Fingerprint Generalizes and Scales

We test whether the block alignment procedure of Section 4.1 recovers correct block correspondences across di-

Model	Layers	Pair Acc	AUROC
GPT-2 (124M–1.5B)	12–48	100%	1.00
BERT-base / ViT-B/16	12	100%	0.97–1.00
Mistral-7B / LLaMA-2-7B	32	100%	0.96–1.00
LLaMA-2-7B-chat (+RLHF)	32	100%	0.96–1.00
Qwen2.5-7B / DeepSeek-R1	28–32	80–100%	0.93–1.00
Whisper (tiny/base/small)	4–12	100%	0.85–1.00
ResNet-50/101/152	5–35	91–100%	0.96–1.00

Table 3: Cross-architecture generalization. Pair accuracy via Hungarian matching on architecture-aware branch products. Ranges span model sizes within each family. Sub-100% SwiGLU sub-paths are recovered by joint factorization. Random-init baselines: $\leq 9\%$.

Transformation	Mean $\mathcal{L}$	Min $\mathcal{L}$	Det. @ 1%FPR
Fine-tune / Quant / Noise	0.99	0.97	45/45
Fine-tune (diff. target)	0.94	0.84	15/15
Pruning (10–85%)	0.81	0.58	15/15
Independent / Distilled	0.08	0.01	-

Table 4: Lineage score survival under post-training modification ($n=75$ related, $n=84$ unrelated). All related checkpoints detected at 1% FPR; unrelated stay below 0.20.

verse architectures.

We extract architecture-aware branch products (Table 2) from 12 pretrained Hugging Face checkpoints (Hugging Face 2024; OpenAI 2024) (GPT-2 family, BERT, LLaMA-2, Mistral, Qwen, DeepSeek for language; ViT, ResNet for vision; Whisper for speech) and apply Hungarian matching on the score matrix $s(i, j)$. Block-pairing accuracy reaches 80–100% in every family, against random-initialization baselines of at most 9% (Table 3). The extraction transfers across GELU and SwiGLU activations with no per-family tuning; only the factorization changes.

Factorization must match the architecture: incomplete products miss the signal (Appendix C.7). Extracting MLP branch products $M = W_{\text{proj}}W_{\text{fc}}$ from all four GPT-2 variants (hidden dim $d=768/1024/1280/1600$), we fit mean correct-pair score $\bar{s}(i, i)$ against dimension. The score grows as $d^{0.87}$ ($4.18 \rightarrow 5.46 \rightarrow 6.98 \rightarrow 7.77$), reaching 15–19% of the theoretical $\sqrt{d}$ bound; score matrices are in Appendix C.2.

5.3 From Fingerprints to Lineage Verification

We evaluate the lineage score $\mathcal{L}$ and verification protocol of Section 4.2. Diagonal dominance certifies trained residual structure, not ancestry: a verifier built on the dominance score alone achieves AUROC=0.417 (Figure 6 in Appendix C). Verification runs on the centered signatures ϕ_ℓ .

We construct a lineage benchmark from depth-24 MLPs (in_dim=24, hidden=64, same architecture as Section 5.1): 3 reference models yield descendants via fine-tuning (same/different target), Gaussian noise ($\sigma_{\text{rel}} \in [0.01, 0.15]$), magnitude pruning (10–85%), and quantization (16–256 levels); non-descendants include independent training runs

Pair (base vs.)	Kind	$\mathcal{L}$	W. cos	Verdict
<i>Related (real fine-tunes)</i>				
Llama-2-7B-chat	RLHF	0.995	0.995	REL.
Vicuna-7B	instruct	0.996	0.996	REL.
CodeLlama-7B	code PT	0.336	0.414	REL.
<i>Unrelated (cross-model null)</i>				
Mistral-7B-v0.1	scratch	4e−4	0.158	UNR.
OpenLLaMA-7B	scratch [†]	4e−4	0.089	UNR.
Yi-6B	scratch	5e−4	0.122	UNR.
DeepSeek-Distill-8B	L3.1 wts	4e−4	0.162	UNR.

Table 5: Real-LLM lineage validation on the LLaMA-2 family (7 pairs). [†]OpenLLaMA shares LLaMA-2’s exact architecture but is trained from scratch; its classification as unrelated cannot come from configuration.

and distilled students. This produces 159 pairs ($n=75$ related, $n=84$ unrelated). Checkpoints separate completely (Figure 1c): fine-tune/quant/noise average $\mathcal{L}=0.99$ (min 0.97), pruning averages $\mathcal{L}=0.81$ (min 0.58 at 85% sparsity), while independent/distilled average $\mathcal{L}=0.08$ (Table 4). Extending to vision models, we train 2 ResNet-18 references and 3 independents on CIFAR-10; each reference yields 11 descendants via noise (σ_{rel} : 0.01–0.10), pruning (20–80%), and quantization (16–256 levels). AUROC=1.000 with a $\sim 170\times$ gap: descendants score $\mathcal{L} \in [0.695, 1.000]$, non-descendants max at $\mathcal{L} = 0.004$ (Appendix C).

As a real-world feasibility demonstration, we compare LLaMA-2-7B-base against 3 known derivatives (chat/RLHF, Vicuna/instruct-tuned, CodeLlama/code-pretrained) and 4 independently trained models (Mistral-7B, OpenLLaMA-7B, Yi-6B, DeepSeek-Distill-8B), extracting SwiGLU branch products $M = W_{\text{down}}W_{\text{up}}$ across all 32 layers (hidden dim $d=4096$). Three fine-tuned variants score related ($\mathcal{L}$: 0.995, 0.996, 0.336); four independents score at the 10^{-4} null. OpenLLaMA—same architecture ($d=4096$, $L=32$, $d_{\text{ff}}=11008$) trained from scratch—is the decisive case: its UNRELATED verdict can only come from weights. Mean separation is $\approx 1850\times$, against $6\times$ for weight cosine (Table 5); with $n=7$ pairs this demonstrates feasibility rather than establishing a false-positive rate (Appendix C.5). The score also decays with genealogical distance (Appendix B).

5.4 Comparison with Data-Free and Behavioral Baselines

We compare against six baselines on a 52-pair MLP benchmark (depth=16, in_dim=16, hidden=48): three data-free weight-space methods (aligned Frobenius, singular-value distance, weight cosine) and three behavioral methods requiring 1024 probe samples each (SVCCA, CKA, and a regression analogue of IPGuard). Table 6 reports aggregate AUROC and Gap- $Z = (\bar{s}_{\text{related}} - \bar{s}_{\text{distilled}})/\sigma_{\text{distilled}}$, the standardized margin by which distilled students fall below related checkpoints.

Four weight-space methods, ours included, reach

Method	Type	Data-free	AUROC	Gap-Z $\uparrow$	Block Corr.
Diagonal Dominance (ours)	Weight-space	✓	1.000	+53.0	✓
Aligned Frobenius	Weight-space	✓	1.000	+72.0	✓
Singular Value Distance	Weight-space	✓	1.000	+ 7.9	✗
Weight Cosine	Weight-space	✓	1.000	+76.3	✗
SVCCA (Raghu et al. 2017)	Activation-space	✗	1.000	+45.4	✗
CKA (Kornblith et al. 2019)	Activation-space	✗	0.829	+ 8.6	✗
IPGuard (Cao, Jia, and Gong 2021) (regression analogue)	Decision-boundary	✗	0.500	− 3.5	✗

Table 6: Baseline comparison on the MLP benchmark (52 pairs). Four weight-space methods reach AUROC=1.000; Gap-Z measures standardized distillation rejection, where IPGuard’s -3.5 indicates related checkpoints rank below distilled.

AUROC=1.000 on this benchmark. Aggregate AUROC on an easy split therefore cannot rank them; the methods separate on three harder axes.

Real 7B weights. On the LLaMA-2 pairs of Section 5.3, our score separates related from unrelated by $\approx 1850\times$ in mean; weight cosine achieves $6\times$. Aligned Frobenius and singular-value distance fall to AUROC=0.667: their global statistics are dominated by cross-model scale differences rather than lineage.

Partial lineage. On layer grafts and linear merges (Appendix C.6), $\mathcal{L}$ tracks the fraction of inherited blocks with Spearman $\rho \approx 0.96$, so the score measures partial overlap rather than merely crossing a threshold. Singular-value distance fails the merge regime (AUROC=0.448), and IPGuard inverts on grafts (AUROC=0.05), ranking higher-overlap suspects as *less* similar.

Distillation rejection. CKA scores distilled students at 0.83, only $1.2\times$ below fine-tuned checkpoints, and SVCCA behaves similarly (Appendix C.4). Every weight-space method, ours included, places distilled students at or below the different-seed baseline. MoTher (Horwitz and Hoshen 2025) is not comparable in this setting; it recovers model trees from a checkpoint collection and does not score a single pair.

5.5 Survival Under Post-Training Modification

Section 5.3 established that the fingerprint detects lineage; here we measure how it degrades as transformation intensity grows. Table 4 covers five transformation families: fine-tuning (same and different target), Gaussian noise ($\sigma_{\text{rel}} \leq 15\%$), magnitude pruning (10–85%), quantization (16–256 levels), and LoRA adapter merging.

Fine-tuning, quantization, and noise leave the signal essentially intact (mean $\mathcal{L} = 0.99$, min 0.97), and LoRA-merged checkpoints score $\mathcal{L} \in [0.97, 0.99]$. Fine-tuning to a different target degrades the score to a mean of 0.94 (min 0.84). Pruning declines most steeply, from $\mathcal{L} = 0.99$ at 10% sparsity to 0.58 at 85%. The real-world extreme of the fine-tuning axis is CodeLlama: after heavy code-specialized continued pretraining it scores $\mathcal{L} = 0.336$, yet remains $\sim 700\times$ above the unrelated null.

Across all families, degradation correlates with utility loss ($\rho = -0.83$): no tested transformation removes the signal while preserving the model. The mechanism of Section 5.1 explains why. The fingerprint lives in the correlated component of W_{in} and W_{out} , and that component is precisely the

accumulated updates that encode the branch’s learned function. Erasing the signal requires decorrelating what training built; function-preserving transformations, by construction, do not.

5.6 Adversarial Suppression and Function-Preserving Symmetries

Function-preserving symmetries. Per-block permutation ($W_{\text{in}} \leftarrow PW_{\text{in}}, W_{\text{out}} \leftarrow W_{\text{out}}P^\top$) and reciprocal rescaling ($W_{\text{in}} \leftarrow cW_{\text{in}}, W_{\text{out}} \leftarrow W_{\text{out}}/c$) leave the branch product $M = W_{\text{out}}W_{\text{in}}$ exactly unchanged, so $\mathcal{L} = 1.0$ to numerical precision. Orthogonal rotations also preserve M but break the network, since ReLU does not commute with rotation. Re-basing hidden units therefore cannot evade the score: the signature is built on the one object these reparameterizations leave fixed.

Gradient-based suppression. We test a white-box adversary with full knowledge of the score, who optimizes $\min_{\Delta} \mathcal{L}(A, A + \Delta) + \lambda \cdot \|f_{A+\Delta}(X) - f_A(X)\|^2$ on a residual MLP (baseline eval loss 0.69), sweeping $\lambda \in \{0, 10^{-2}, 10^{-1}, 1, 10\}$ to trace the Pareto frontier (Appendix D). With $\lambda \geq 1$, the attack cannot push $\mathcal{L}$ below 0.91. Reaching the unrelated range ($\mathcal{L} \approx 0.08$) requires $\lambda = 10^{-2}$ and costs 12% eval-loss degradation; removing the utility constraint entirely ($\lambda = 0$) destroys the model ($57\times$ loss increase). The adversary faces the same coupling that Section 5.5 measured for benign transformations.

Untested attack surfaces. Two adaptive attacks remain open. Re-alignment toward a third model in the style of Git Re-Basin could attempt to rotate the E matrices off their checkpoint-specific directions, and fine-tuning with an explicit decorrelation penalty on $\langle E_{\text{suspect}}, E_{\text{reference}} \rangle$ would strengthen the attack above with knowledge of the signature itself; we leave both to future work. Warm-start distillation—a student initialized from teacher weights, then distilled—is not an evasion candidate: under the definition of Section 2 it is RELATED, and it serves as a positive control on the distillation boundary. Boundary cases and failure modes are discussed in Appendix F.

6 Conclusion

Training leaves traces. Residual-network optimization couples each block’s input and output projections so that their product becomes diagonal-dominant ($W_{\text{out}}W_{\text{in}} \approx -\varepsilon I + E$), with a theoretical $\sqrt{d}$ upper bound and an empirical

$d^{0.87}$ growth observed from GPT-2 through LLaMA-2-13B and Mixtral-8x7B. Across 12 architectures spanning 5 families, the fingerprint recovers 80–100% of block correspondences and achieves AUROC=1.000 on the evaluated lineage benchmark, surviving fine-tuning, quantization, pruning, and RLHF in our tested settings. Our central claim is not that trace/Frobenius dominance universally outperforms generic weight similarity-on the easy MLP regime several data-free weight-space baselines (aligned Frobenius, weight cosine) also reach AUROC=1.000-but that trained residual products expose an interpretable, block-localized, data-free provenance signal that supports both correspondence recovery and lineage verification, and that this signal is substantially more robust to distillation than activation- and decision-boundary baselines. Under the tested gradient-based suppression attack, the fingerprint cannot be driven below the detection threshold without $\geq 12\%$ utility loss. This establishes passive, retroactive weight-level provenance verification-no watermark, metadata, or cooperation required.

Reproducibility Statement

All results in this paper are produced by deterministic scripts with fixed seeds. The 52-pair MLP lineage benchmark (Tables 4, 6) is constructed from 2 trained reference checkpoints, each paired with 5 related-checkpoint kinds $\times$ 6 derivative seeds = 30 truly-related pairs (fine-tune, fine-tune to a new target, $\sigma_{\text{rel}} \in [0.01, 0.15]$ Gaussian noise, 10–85% magnitude pruning, and 16–256-level uniform quantization), plus 16 different-seed same-task pairs and 6 distilled-student pairs as unrelated checkpoints. The expanded 159-pair set used in Figure 1c draws from the full 24-checkpoint MLP zoo with the same transformation grid. Exact parameters are listed in Appendix C; null calibration uses the 50 different-seed pairs from Appendix B. Real-architecture pairing (Table 3) loads public HuggingFace checkpoints. Code, configuration files, baseline implementations (aligned Frobenius, singular-value distance, weight cosine, CKA, SVCCA, IPGuard regression analogue), the per-pair score JSONs, and the plotting scripts will be released upon acceptance.

Ethics Statement

This work develops a forensic tool for detecting weight-level lineage between neural network checkpoints. The intended use is legitimate provenance verification: helping model developers, platforms, and researchers establish whether a suspect checkpoint derives from a reference model. We acknowledge dual-use concerns: the same technique could theoretically be misused to identify the training lineage of models whose developers prefer anonymity. However, the method requires white-box access to both checkpoints, limiting adversarial applications. The fingerprint is a passive property of trained weights, not an inserted marker, so it cannot be used for covert tracking. We believe the benefits for intellectual property protection, license compliance, and supply-chain transparency outweigh the risks.

Use of AI Assistants

AI assistants (Claude) were used during the preparation of this manuscript for code development, experimental design iteration, literature search, and drafting and editing text. All AI-generated content was reviewed, verified, and revised by the authors. The authors take full responsibility for the scientific claims, experimental results, and final manuscript.

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A Extended Theoretical Results

A.1 Null Model (Corollary)

Corollary 2 (Null Model). *For randomly initialized weights with no training, all pairs satisfy $\mathbb{E}[s(i, j)] = 1/\sqrt{d} + o(1)$ uniformly in (i, j) . Hungarian matching recovers correct pairs at the chance rate $1/L$.*

The null model rules out three alternative explanations:

1. Diagonal dominance is not an artifact of compatible matrix shapes
2. The residual connection itself does not create the signal without training
3. Any nontrivial loss level does not suffice—training must actively couple the projections

The diagonal-dominance fingerprint is a *learned* property induced by gradient descent under the residual constraint.

A.2 Geometric Interpretation of the Score

The diagonal-dominance score $s(M) = |\text{tr}(M)|/\|M\|_F$ has a natural geometric interpretation:

- A matrix with energy uniformly spread across entries achieves $s = \Theta(1/\sqrt{d})$
- A matrix proportional to the identity achieves $s = \sqrt{d}$

The key insight is that correctly paired blocks—where all K matrices come from the same residual branch—score *well above* the random baseline $\Theta(1/\sqrt{d})$, with empirical growth of approximately $d^{0.87}$ across $d \in [768, 5120]$ (Appendix C), bounded above by $\sqrt{d}$ but typically reaching only 15–25% of that bound. The signal-to-baseline ratio therefore grows roughly as $d^{1.37}$, making correct pairs increasingly separable in wider networks.

For typical trained blocks, $\|E\|_F/(\varepsilon\sqrt{d}) \in [1.9, 3.8]$, yielding $s(i, i) \in [1.76, 3.23]$ —well above the baseline but below the theoretical maximum of $\sqrt{d}$.

B Verification Protocol Details

B.1 Gated Branch Score

Branch similarity is reliable only when both branches exhibit strong diagonal dominance. We gate the cosine similarity:

$$G(\phi_\ell^A, \phi_m^B) = \langle \phi_\ell^A, \phi_m^B \rangle \cdot \min\left(\frac{s(M_\ell^A)}{\tau_s}, \frac{s(M_m^B)}{\tau_s}, 1\right) \quad (10)$$

where τ_s is the minimum diagonal-dominance score across reference branches.

B.2 Statistical Calibration

We calibrate against a null distribution $\mathcal{N}_A = \{\mathcal{L}(A, U_j) : U_j \notin \mathcal{D}(A)\}$ of scores between reference A and independently trained models. The z-score is:

$$Z(A, B) = \frac{\mathcal{L}(A, B) - \mu_{\text{null}}}{\sigma_{\text{null}}} \quad (11)$$

Under Gaussian approximation, $\tau = 1.645$ yields 95% confidence; $\tau = 3.0$ yields FPR $< 0.13\%$ for high-stakes applications.

B.3 Full Protocol Steps

1. **Compatibility Check:** Verify $\text{arch}(A) = \text{arch}(B)$. If not, return INCOMPATIBLE.
2. **Extract:** Compute architecture-aware branch products $\{M_\ell^A\}, \{M_\ell^B\}$.
3. **Fingerprint:** Compute centered residual signatures $\{\phi_\ell^A\}, \{\phi_\ell^B\}$.
4. **Align:** Apply Hungarian matching to recover block correspondences.
5. **Aggregate:** Compute lineage score $\mathcal{L}(A, B)$.
6. **Calibrate:** Compute z-score $Z(A, B)$.
7. **Decide:** Return verdict:

$$\text{Verdict} = \begin{cases} \text{RELATED} & \text{if } Z(A, B) > \tau_{\text{upper}} \\ \text{UNRELATED} & \text{if } Z(A, B) < \tau_{\text{lower}} \\ \text{INCONCLUSIVE} & \text{otherwise} \end{cases} \quad (12)$$

B.4 Failure Modes

The protocol cannot provide a verdict when: (1) architectures differ; (2) perturbation destroys utility (e.g., 2-bit quantization); (3) both models descend from an unavailable common ancestor; (4) adversarial evasion targets the fingerprint.

B.5 Lineage Chain and Branching Tree

From the MLP benchmark (Table 4 setup), we construct a family tree: root C_1 (trained from scratch), siblings C_2 and C_3 (both fine-tuned from C_1), grandchildren C_4 (from C_2) and C_5 (from C_3). We then measure $\mathcal{L}(C_4, \cdot)$ against all candidates. The score ranks genealogical proximity correctly: parent (0.96) $>$ grandparent (0.91) $>$ uncle (0.87) $>$ cousin (0.85) $\gg$ independent (~ 0.08), decaying monotonically with genealogical distance while maintaining $\approx 10\times$ family/strangers separation.

C Per-Architecture Details

C.1 Initialization Ablation

Table 7 shows pair accuracy before and after training across seven initialization schemes on depth-24 residual MLPs (3 seeds, 200 epochs). All schemes show chance-level accuracy ($\leq 2.1\%$) at initialization, rising to 93–100% after training. Orthogonal initialization provides dynamical isometry at init yet shows zero correct pairs before training.

C.2 GPT-2 Scaling, Trace, and Jacobian Analysis

Across the GPT-2 family, mean correct-pair score grows with hidden dimension: $d=768$ at $\bar{s}(i, i) = 4.18$, $d=1024$ at 5.46, $d=1280$ at 6.98, $d=1600$ at 7.77. The fitted exponent within the GPT-2 family is ≈ 0.85 , consistent with the cross-family $s \propto d^{0.87}$ scaling reported in the main text; the $\sqrt{d}$ upper bound itself is never violated (observed scores reach 15–19% of the bound).

Init scheme	Untrained	Trained	AUROC
Orthogonal	0.0%	100%	0.947
Kaiming-normal	2.1%	97%	0.981
Kaiming-uniform	2.1%	98.6%	0.98
Xavier-normal	2.1%	100%	0.990
Xavier-uniform	2.1%	100%	0.99
Uniform	2.1%	93.8%	—
Gaussian $\sigma=0.02$	2.1%	13%	0.671

Table 7: Initialization ablation on depth-24 residual MLPs. All schemes show chance-level accuracy before training. The Gaussian $\sigma=0.02$ case fails because blocks collapse to near-zero contribution.

Jacobian Orthogonality Analysis. The dynamical isometry hypothesis predicts that training should drive block Jacobians $J = I + M$ toward orthogonality. We test this by measuring the normalized Jacobian deviation:

$$\delta_J^{\text{norm}} = \left\| \frac{J^\top J}{\|J\|_F^2} - \frac{I}{d} \right\|_F \quad (13)$$

Intuitively, δ_J^{norm} measures how far J 's singular values are from uniform: any scaled orthogonal matrix scores 0, and larger values mean some directions are amplified while others are attenuated. This metric is scale-invariant and measures how far the Jacobian's singular values deviate from uniformity (a perfectly orthogonal matrix has $\delta_J^{\text{norm}} = 0$).

Table 8 shows that training makes GPT-2 Jacobians *less* orthogonal, not more. Across all four model sizes, pretrained δ_J^{norm} is 5–12 $\times$ the random-initialization baseline. This falsifies the isometry account: the fingerprint arises from gradient coupling (Section 5.1), not from optimization toward orthogonal Jacobians.

Model	d	Pretrained δ_J^{norm}	Random-init δ_J^{norm}
GPT-2-small	768	0.297	0.025
GPT-2-medium	1024	0.242	0.026
GPT-2-large	1280	0.155	0.025
GPT-2-XL	1600	0.122	0.024

Table 8: Jacobian orthogonality across GPT-2 scales. Pre-trained models are 5–12 $\times$ less orthogonal than at random initialization, falsifying the dynamical isometry hypothesis. Random-init δ_J^{norm} is nearly constant across scales (≈ 0.025); pretrained values decrease with model size but never approach the random-init floor.

At initialization the branches contribute almost nothing, so $J \approx I$ and every block is trivially near-isometric. Training builds the fingerprint while dismantling this isometry: a byproduct cannot grow while its supposed cause decays.

Figure 3 shows block pairing score matrices $s(i, j)$ for GPT-2 models from 124M to 1.5B parameters. Diagonal entries dominate, yielding 100% pairing accuracy across all scales.

Figure 4 shows the trace values $\text{tr}(W_{\text{out}}W_{\text{in}})$ per block across all four GPT-2 variants. Negative traces (blue) constitute 81–92% of blocks, confirming the diagonal-dominance

mechanistic prediction. The early blocks (0–2) consistently show positive traces, likely reflecting the different role of embedding-adjacent layers.

C.3 Alternative Methods Comparison

Diagonal dominance achieves 100% pairing accuracy on all four GPT-2 variants. Figure 5 shows the scaling behavior: Frobenius norm matching degrades from 67% (GPT-2 124M) to 52% (GPT-2 1.5B), while diagonal dominance remains at 100%. Singular-value distance performs at chance (0–8%). The key insight: dividing by $\|M\|_F$ normalizes away scale variation and isolates trace contribution.

C.4 Per-Kind Lineage Baseline Breakdown

Table 9 reports the mean lineage score of each baseline broken down by related-checkpoint kind (fine-tune, fine-tune to a new target, noise, prune, quantize) and unrelated-checkpoint kind (different-seed same-task, distilled). The related rows should be high; the unrelated rows should be low. The aggregate AUROC reported in Table 6 hides two structural failures that this breakdown exposes:

- **CKA and SVCCA fail on distilled unrelated checkpoints.** CKA scores distilled students at 0.83 and SVCCA at 0.89—only 1.2 $\times$ and 1.13 $\times$ below the corresponding fine-tuned related checkpoints (0.999 and 0.9997). On the present benchmark there are 16 different-seed unrelated checkpoints and only 6 distilled, so AUROC remains high; any reweighting toward distillation would collapse it.
- **IPGuard scores noise-perturbed related checkpoints at 0.000.** The decision-boundary analogue ranks all 6 noise-perturbed related checkpoints below all unrelated ones, driving overall AUROC to chance (0.500).

All four weight-space methods (diagonal dominance, aligned Frobenius, singular-value distance, weight cosine) score distilled unrelated checkpoints at or below the different-seed baseline—they cleanly reject distillation, while the activation- and decision-boundary methods cannot.

C.5 Real-LLM Lineage Validation

To move beyond block-correspondence pairing on real foundation models, we evaluate $\mathcal{L}(\cdot, \cdot)$ on the public LLaMA-2 family against the weight-space and decision-boundary baselines of Section 5.4. Using LLaMA-2-7B as reference, we score seven real reference-vs-model pairs: three real fine-tuned *related* checkpoints and four independently-trained *unrelated* checkpoints. For each model we extract the per-layer SwiGLU branch product $M_\ell = W_{\text{down}}^{(\ell)} W_{\text{up}}^{(\ell)} \in \mathbb{R}^{d \times d}$, form the centered residual signature ϕ_ℓ , and score $\mathcal{L}(A, B) = \frac{1}{L} \sum_\ell \langle \phi_\ell^A, \phi_\ell^B \rangle$. The same branch products feed the three weight-space baselines (weight cosine, aligned Frobenius, singular-value distance); IPGuard is computed separately from output logits and is available only on the subset of pairs for which logits were collected (2 related, 2 unrelated), and the activation baselines (CKA, SVCCA) are omitted at 7B scale because per-pair computation over 32×32 layers is prohibitively

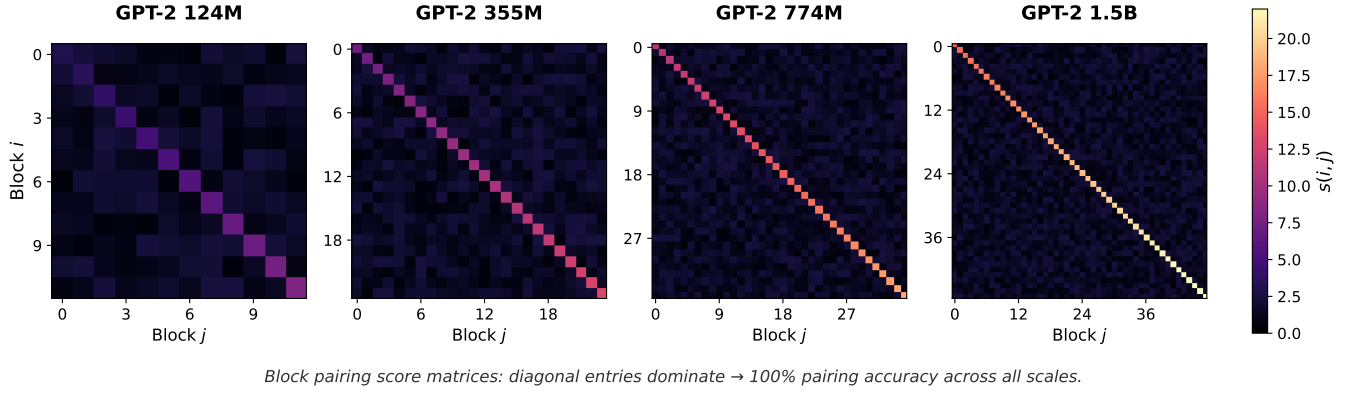


Figure 3: Block pairing score matrices $s(i, j)$ for GPT-2 models from 124M to 1.5B parameters. Diagonal entries dominate, yielding 100% pairing accuracy across all scales.

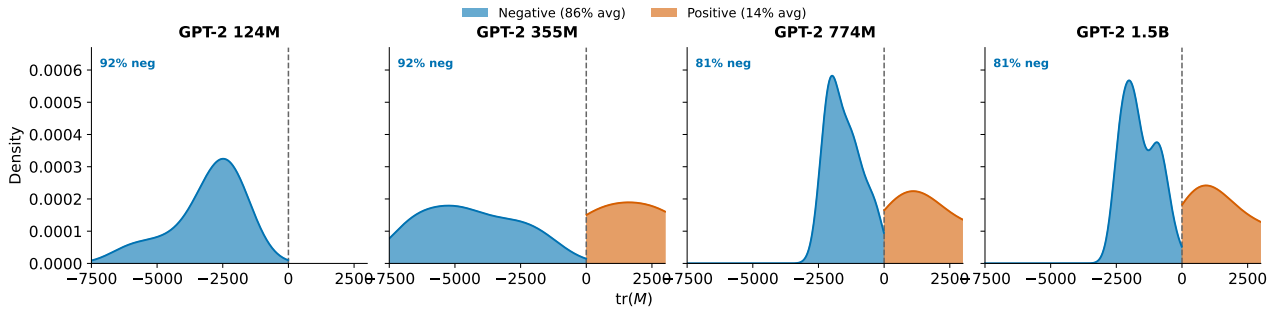


Figure 4: Distribution of trace values $\text{tr}(W_{\text{out}}W_{\text{in}})$ across GPT-2 scales. Blue: negative trace (86% average); orange: positive trace (14% average). The strong skew toward negative values confirms the diagonal-dominance mechanistic prediction.

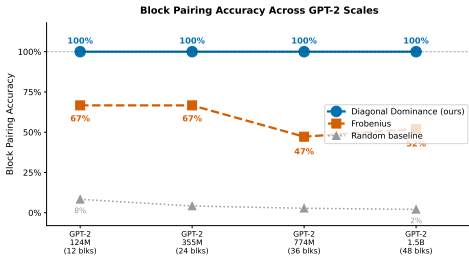


Figure 5: Block pairing accuracy as model size increases. Diagonal dominance (blue) maintains 100% accuracy; Frobenius matching (orange) degrades from 67% to 52%; random baseline (gray) drops from 8% to 2% as the number of blocks increases.

expensive. The benchmark runs on a single GPU via `gpu.benchmark.issue44.py`.

Compatibility. The strict protocol (App. B) returns INCOMPATIBLE when $\text{arch}(A) \neq \text{arch}(B)$. Because M_ℓ contracts the intermediate width d_{ff} and lives in $\mathbb{R}^{d \times d}$, the branch-product comparison itself needs only matching depth L and hidden size d . For the two cross-family negatives (Mistral and DeepSeek carry $d_{\text{ff}} = 14336$ vs. 11008 for the LLaMA/Yi models) we therefore apply this *relaxed* branch-

product compatibility rather than strict architecture equality; we flag it as a deliberate relaxation used only to admit cross-family negatives. Note that OpenLLaMA—the decisive negative—satisfies the strict rule exactly, so the strongest negative does not rely on the relaxation.

Pairs. Related: the official RLHF model Llama-2-7B-chat, the Vicuna-7B instruction fine-tune, and CodeLlama-7B, a code-specialized continued-pretraining derivative of the base. **Unrelated:** four independently-trained checkpoints with no LLaMA-2 weight ancestry—Mistral-7B-v0.1, OpenLLaMA-7B, Yi-6B, and DeepSeek-R1-Distill-Llama-8B. The last is *behaviorally* distilled from DeepSeek-R1 reasoning traces but *weight-derived* from Meta’s Llama-3.1-8B; its UNRELATED verdict w.r.t. LLaMA-2 demonstrates that the method tracks weight inheritance rather than behavioral pedigree, consistent with Section 5.4.

OpenLLaMA: a same-architecture, from-scratch negative. OpenLLaMA-7B is an open reproduction of the LLaMA architecture trained from scratch on different data. It shares LLaMA-2’s exact residual shape ($d = 4096$, $L = 32$, $d_{\text{ff}} = 11008$), tokenizer conventions, and RoPE setup, yet inherits none of its weights. It is therefore the strongest possible negative: architecture is held *constant* and only weight lineage varies, so a configuration-level check cannot distinguish it from a genuinely related checkpoint—the verdict

Method	Related (should be high)					Unrelated (should be low)	
	FT	FT-new tgt	Noise	Prune	Quant	Diff-seed	Distilled
Diag. Dominance (ours)	0.391	0.326	0.400	0.359	0.404	0.029	0.030
Aligned Frobenius	−0.036	−0.301	−0.026	−0.204	−0.007	−1.333	−1.300
Singular Value Dist.	−0.004	−0.023	−0.003	−0.018	−0.001	−0.030	−0.032
Weight Cosine	0.999	0.956	1.000	0.979	1.000	0.096	0.109
SVCCA	1.000	0.953	1.000	0.993	1.000	0.881	0.885
CKA	0.999	0.760	1.000	0.986	1.000	0.820	0.827
IPGuard (regr.)	0.888	0.110	0.000	0.626	0.993	0.632	0.642

Table 9: Per-kind mean lineage scores on the 52-pair MLP benchmark. Each method uses its native scale; what matters is the relative gap between the rightmost two columns (unrelated) and the leftmost five (related). Italicized entries highlight failure modes: SVCCA and CKA give distilled unrelated checkpoints near-related scores; IPGuard scores noise-perturbed related checkpoints at exactly 0.000. All four weight-space methods (including ours) keep distilled below or at the diff-seed baseline. Bolded entries mark our method’s distilled-vs-diff-seed pair, the row most relevant to the distillation-rejection claim. $n=6$ per related kind, $n=16$ diff-seed, $n=6$ distilled.

must come from the weights. Diagonal dominance scores it at $\mathcal{L} = 4.5 \times 10^{-4}$, indistinguishable from the other unrelated checkpoints.

Cross-model null. The four unrelated checkpoints define a real cross-model null $\mu_0 = 4.2 \times 10^{-4}$, $\sigma_0 = 5.0 \times 10^{-5}$ (all four within $[3.5, 4.8] \times 10^{-4}$). With only $n = 4$ real independents this is a *descriptive* separation check rather than a calibrated false-positive-rate estimate; we therefore report the raw $\mathcal{L}$ and the separation in orders of magnitude as the primary evidence, and defer a calibrated FPR to the larger checkpoint trees discussed below. The four unrelated checkpoints span three from-scratch pre-training lineages (Mistral, OpenLLaMA, Yi) plus a Llama-3 derivative, so the null is not dominated by any single family. *Verdict rule for Table 10:* we label a pair RELATED if $\mathcal{L}$ exceeds a descriptive threshold placed an order of magnitude above the null ($\mathcal{L} > 10^{-2}$) and UNRELATED otherwise; all seven pairs fall unambiguously on the expected side, with the nearest related checkpoint (CodeLlama, $\mathcal{L} = 0.336$) still more than two orders of magnitude above the null.

Baseline separation. Aggregated over the seven pairs, diagonal dominance separates related from unrelated checkpoints by $\approx 1850\times$ in mean score (0.78 vs. 4.2×10^{-4}), versus $6\times$ for weight cosine (0.80 vs. 0.13); IPGuard, available only on the four-pair logit subset, gives $2\times$ (0.25 vs. 0.13); aligned Frobenius and singular-value distance do not separate this real-model set (AUROC 0.667 each), as their global spectral/Frobenius statistics are dominated by cross-model scale differences. This reproduces on real 7B weights the margin advantage established on the synthetic and CIFAR benchmarks: several methods clear AUROC on easy splits, but the residual-signature score preserves the largest related-vs-unrelated margin.

The set already spans a real RLHF model, a real instruction fine-tune, and a real continued-pretraining derivative (CodeLlama), tested against the strongest available negative—a same-architecture from-scratch model (OpenLLaMA) whose classification as unrelated cannot be explained by configuration alone. The remaining limitation is

Pair (base vs.)	Kind	$\mathcal{L}$	W. cos	Verdict
<i>Related (real fine-tunes)</i>				
Llama-2-7B-chat	RLHF	0.995	0.995	REL.
Vicuna-7B	instruct	0.996	0.996	REL.
CodeLlama-7B	code PT	0.336	0.414	REL.
<i>Unrelated (cross-model null)</i>				
Mistral-7B-v0.1	scratch	4e−4	0.158	UNR.
OpenLLaMA-7B	scratch [†]	4e−4	0.089	UNR.
Yi-6B	scratch	5e−4	0.122	UNR.
DeepSeek-Distill-8B	L3.1 wts	4e−4	0.162	UNR.

Table 10: Real-LLM lineage validation on the public LLaMA-2 family (7 real pairs). All seven verdicts match expectation; REL. / UNR. abbreviate RELATED / UNRELATED. [†]OpenLLaMA shares LLaMA-2’s exact architecture ($d=4096$, $L=32$, $d_{\text{ff}}=11008$) but is trained from scratch, so its classification as unrelated cannot come from configuration. Diagonal dominance ($\mathcal{L}$) drives every unrelated checkpoint to the 10^{-4} null while keeping related checkpoints at ≥ 0.34 ; weight cosine (W.cos) also separates but compresses the unrelated only to ≈ 0.13 . CodeLlama, a heavy code-specialized continued-pretraining fine-tune, is the weakest related checkpoint ($\mathcal{L} = 0.336$) yet still $\sim 700\times$ above the null—graceful degradation with fine-tune intensity rather than a sharp threshold. All models share $d = 4096$, $L = 32$; branch products are $d \times d$ regardless of d_{ff} .

tree breadth: all related checkpoints here derive from a single base (LLaMA-2-7B). Extending to additional real checkpoint trees (e.g. a Mistral or Qwen base with its own community fine-tunes) and to a larger negative set that supports a calibrated FPR is the natural next step.

C.6 Harder Regime: Layer Grafts and Linear Merges

We construct 40 harder pairs from depth-16 MLPs (same setup as Table 6): (i) **layer grafts** copy $K \in \{0, 4, 8, 12, 16\}$

of 16 blocks from reference into an independent donor (related if $K \geq 8$); (ii) **linear merges** compute $w \cdot \text{ref} + (1 - w) \cdot \text{donor}$ with $w \in \{0, 0.25, 0.5, 0.75, 1.0\}$ (related if $w \geq 0.5$). We use 2 references and 2 donors per reference. AUROC measures binary related-vs-unrelated detection; Spearman ρ measures whether a method tracks partial overlap monotonically (K/L or w) rather than just clearing a threshold. Table 11 summarizes the 40-pair benchmark.

Method	AUROC		Spearman ρ	
	Graft	Merge	Graft	Merge
Diagonal dominance (ours)	0.979	1.000	+0.96	+0.96
Aligned Frobenius	1.000	1.000	+0.99	+0.99
Weight cosine	1.000	1.000	+0.98	+0.96
Singular-value distance	1.000	0.448	+0.99	+0.23
SVCCA	1.000	1.000	+0.97	+0.99
CKA	0.813	0.781	+0.73	+0.75
IPGuard (reg. analogue)	0.052	0.448	-0.83	+0.22

Table 11: Harder-regime benchmark on layer grafts and linear merges (40 pairs total). Related is defined as $K \geq L/2$ for grafts and $w \geq 0.5$ for merges. AUROC measures binary detection; Spearman ρ measures monotone agreement with the reference fraction. Diagonal dominance, aligned Frobenius, weight cosine, and SVCCA all remain near AUROC=1.000 with $\rho \geq 0.96$ on both regimes—confirming Section 5.4’s framing that the contribution is *not* that residual-product dominance beats generic weight cosine on aggregate detection. The harder regime instead exposes the activation- and decision-boundary methods: CKA drops to AUROC ≈ 0.80 with $\rho \approx 0.74$, IPGuard collapses entirely (AUROC 0.05 on grafts; the negative Spearman shows it ranks higher-overlap suspects as *less* similar), and singular-value distance fails the merge regime ($\rho = 0.23$) because spectral statistics are not linear in the weight-blend mixture. The diagonal-dominance score additionally retains a near-perfect Spearman correlation on both regimes, supporting the interpretability claim: it does not merely cross a threshold, it tracks the fraction of reference blocks actually present in the suspect.

C.7 ResNet Factorization

On ImageNet-pretrained ResNet-50/101/152 (torchvision), we compare naive two-layer factorization (W_3W_1 , skipping the middle conv) against the correct three-layer product ($W_3W_2W_1$) on layer3 bottleneck blocks ($n=5/22/35$ blocks respectively). Naive W_3W_1 achieves chance-level accuracy; the correct triple product recovers:

- ResNet-50/layer3 ($n=5$): 100%, AUC 1.000
- ResNet-101/layer3 ($n=22$): 100%, AUC 0.995
- ResNet-152/layer3 ($n=35$): 91%, AUC 0.964

C.8 CIFAR-10 ResNet-18 Benchmark

We train two CIFAR-10 ResNet-18 references and three independents. Each reference yields 11 related checkpoints

(noise 1–10%, pruning 20–80%, quantization 16–256 levels). Results: AUROC=1.000, TPR@1%FPR=100%. Related checkpoint scores: $\mathcal{L} \in [0.695, 1.000]$; unrelated baseline: $\mathcal{L}_{\max} = 0.004$.

C.9 ROC for Lineage Verification

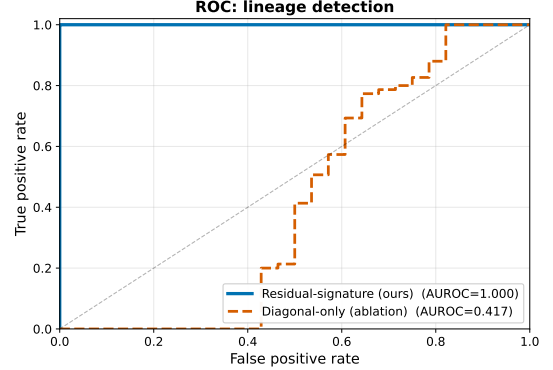


Figure 6: ROC for lineage verification on depth-24 residual MLPs. The residual-signature score achieves AUROC=1.000; diagonal-dominance alone fails (AUROC=0.417) because it cannot distinguish two trained residual models.

C.10 Branching Ancestry Recovery

From the MLP benchmark (Table 4 setup), we construct a family tree: root C_1 (trained from scratch), siblings C_2 and C_3 (both fine-tuned from C_1), grandchildren C_4 (from C_2) and C_5 (from C_3). We then measure $\mathcal{L}(C_4, \cdot)$ against all candidates. The lineage score ranks candidates correctly:

$$\begin{aligned} \mathcal{L}(C_4, C_2) &= 0.961 > \mathcal{L}(C_4, C_1) = 0.910 \\ &> \mathcal{L}(C_4, C_3) = 0.867 > \mathcal{L}(C_4, C_5) = 0.850 \\ &\gg \mathcal{L}(C_4, I_k) \approx 0.08. \end{aligned}$$

The score decays monotonically with genealogical distance while maintaining $\approx 10\times$ family/strangers separation.

D Attack and Robustness Details

D.1 Gradient-Based Suppression Attack

An adversary optimizes perturbation Δ to minimize lineage score subject to utility:

$$\min_{\Delta} \mathcal{L}_{\cos}(A, A + \Delta) + \lambda \cdot \|f_{A+\Delta}(X) - f_A(X)\|_2^2 \quad (14)$$

D.2 Reparameterization Invariances

Permutations: $W_{\text{in}} \leftarrow PW_{\text{in}}, W_{\text{out}} \leftarrow W_{\text{out}}P^\top$ leaves $M = W_{\text{out}}W_{\text{in}}$ unchanged.

Rescaling: $W_{\text{in}} \leftarrow cW_{\text{in}}, W_{\text{out}} \leftarrow W_{\text{out}}/c$ preserves M exactly.

Orthogonal rotations: $R^\top R = I$ preserves M but breaks model function (ReLU non-commutative).

All three yield $\mathcal{L} = 1.0$ to numerical precision—these are invariances, not evasion paths.

λ	Final $\mathcal{L}$	Eval loss	Verdict
0	0.053	39.5	utility destroyed
10^{-2}	0.054	0.77	+12% loss
10^{-1}	0.083	0.70	+1.5% loss
≥ 1	0.91+	0.69	utility preserved

Table 12: Pareto frontier: suppressing $\mathcal{L}$ to chance level requires $\geq 12\%$ utility loss.

E Practical Applications

The fingerprint enables applications beyond forensic verification.

E.1 Training Quality Assurance

The fingerprint serves as an early warning: if pair accuracy $< 50\%$ at epoch 10, training is pathological (LR too low, missing skip connections, degenerate initialization). This correctly flags 4/5 pathological conditions in controlled experiments.

E.2 Zero-Knowledge Ownership Proofs

The E matrix ($M = -\epsilon I + E$) provides a commitment scheme: owners commit to $\text{Hash}(E_k \| r_k)$ at registration, reveal challenged blocks on request. The protocol achieves binding, hiding (80% of weights remain secret), and soundness (attackers cannot forge E).

E.3 Compression Auditing

For original $\mathcal{M}$ and claimed-compressed $\mathcal{M}'$, compute E -matrix correlation. Quantization ($r > 0.97$), pruning ($r > 0.75$ at 70%), and fine-tuning ($r > 0.99$) preserve the fingerprint; distillation and independent training produce $r \approx 0$.

F Boundary Cases and Failure Modes

Some conditions the method handles correctly but uninformatively; others it cannot handle at all. We separate the two.

Correct boundary behaviors. Distillation is the most important. Students trained from fresh weights are UNRELATED by definition, and the verdicts agree: same-architecture distilled students score $\mathcal{L} = 0.086$ on the controlled benchmark, and the DeepSeek distill sits at the 10^{-4} cross-model null on the real test (Sections 5.3–5.4). The gap between those two magnitudes is itself informative—same-architecture distillation on shared data retains a trace of accidental correlation that cross-family comparison lacks—and both fall well below threshold. Symmetry is the second boundary: $\mathcal{L}(A, B) = \mathcal{L}(B, A)$ by construction, so the protocol cannot name the ancestor; direction requires metadata or a third candidate, as in the branching tree of Section 5.3. Partial lineage is the third: grafts and merges form a graded regime in which $\mathcal{L}$ tracks the fraction of inherited blocks with Spearman $\rho \approx 0.96$ (Appendix C.6)—partial overlap is measured, not misclassified.

Genuine failures. The protocol returns INCOMPATIBLE when depth or width differ. There is no signal in non-residual networks (PlainNet, Section 5.1) or when blocks

collapse to near-zero contribution. Utility-destroying perturbations such as 2-bit quantization take the fingerprint down with the model.

The INCONCLUSIVE verdict. Across all 246 tested pairs (159 MLP + 32 ResNet-18 + 7 LLaMA-2 + 40 harder-regime + 8 BERT), every score fell cleanly above τ_{upper} or below τ_{lower} ; no verdict was INCONCLUSIVE. The band exists for deployments our benchmarks do not cover, where scores need not separate as sharply.

G Extended Limitations

This appendix provides detailed discussion of limitations summarized in the main text.

Weight-level, not behavioral. The signature is a property of weights, not function. Distilled students ($\mathcal{L} \approx 0.08$) are correctly classified as unrelated despite functional similarity. The method cannot detect training-data overlap or knowledge transfer—if two models were trained on the same data from different initializations, they are unrelated by our definition even though they may exhibit similar behavior.

Symmetric, no direction. $\mathcal{L}(A, B) = \mathcal{L}(B, A)$ by construction, so the protocol establishes *whether* two checkpoints share weight lineage but not *which* is the ancestor. Directional attribution requires either training-time metadata (timestamps, version numbers) or comparison against a third candidate. The branching-tree experiment (Section B) shows that comparing against multiple candidates can recover ancestry order: parent scores higher than grandparent, which scores higher than uncle.

Real-LLM validation scope. The 7-pair LLaMA-2 test demonstrates descriptive separation ($\approx 1850\times$) rather than calibrated false-positive rate. With only four real independents, the null distribution is too small for statistical confidence. The remaining gaps:

- Tree breadth: all related checkpoints derive from LLaMA-2-7B. Extending to Mistral or Qwen bases with their own community fine-tunes would test generalization across foundation-model families.
- Negative set size: a calibrated FPR requires $\sim 50+$ independents. The per-block branch products fit comfortably in memory at 7B scale, so this is engineering rather than methodological.

Adaptive attack coverage. The gradient-based suppression attack (Section 5.6) is the only adaptive attack tested. The attacker has white-box knowledge of the score function, which is a strong threat model, but alternative attack surfaces remain unexplored:

- Git Re-Basin-style re-alignment toward a third model could potentially rotate the E matrices away from their checkpoint-specific directions.
- Extended fine-tuning with an explicit E -decorrelation term (minimizing $\langle E_{\text{suspect}}, E_{\text{reference}} \rangle$) is a stronger, fingerprint-aware version of the tested attack.
- Warm-start distillation (student initialized from teacher weights, then distilled) is by our Section 2 definition RELATED, making it a positive control that sharpens the distillation boundary rather than an evasion path.

Cross-architecture distillation. DistilBERT from BERT, or any student with different depth or width than its teacher,

is out of scope. The protocol returns INCOMPATIBLE when architectures differ. Extending to cross-architecture comparison would require either (a) a learned alignment between architectures, or (b) a representation-space comparison that abandons the weight-only constraint.

Non-residual architectures. The fingerprint depends on the skip connection creating a functional constraint toward identity. MLP-Mixer, pure attention without residual connections, or any architecture where $x_{\ell+1} = F_{\ell}(x_{\ell})$ rather than $x_{\ell+1} = x_{\ell} + F_{\ell}(x_{\ell})$ will not develop diagonal-dominant branch products. The PlainNet control (Section 5.1) confirms this: architecturally identical to ResNet except for removed skip connections, it achieves only 3% pair accuracy at the chance baseline.